Radiology AI | Web, mobile and accessibility strategy

Frontend specification package - October 2026 | Planning baseline, not a repository audit

Web/mobile experience

Design principles - Clinical-first hierarchy, calm neutral palette, clear status badges, readable typography, keyboard navigation, large touch targets, dark mode only after clinical usability testing. Never use color alone to signal critical status.

Navigation - Left sidebar for core modules, persistent top bar with institution/context, breadcrumbs, patient/study context header, global search with authorization-aware results.

Accessibility - Target WCAG 2.2 AA for app chrome; test screen-reader labels, focus management, zoom, contrast, keyboard shortcuts and screen-size constraints. Viewer-specific accessibility gaps must be explicitly documented.

Internationalization - Albanian and English localization from the outset, locale-aware date/number formats, 24-hour clock option, units displayed explicitly, medically reviewed terminology and no automatic translation of clinical conclusions.

Responsive strategy - Desktop-first clinical workspaces (1440px+); usable tablet layouts for worklists and review; mobile focuses on safe read-only tasks and notifications, not complex primary diagnosis.

Interaction rules - Always distinguish draft, unreviewed AI suggestion, clinician-reviewed and signed reports. Show provenance, model/version, timestamp and supporting source when available; allow correction and rejection with audit trail.

Mobile deployment choices

Phase 1 - Responsive React web app for worklists and non-diagnostic workflows.

Phase 2 - Evaluate React Native app only if native notifications, secure device features or dedicated workflows justify maintenance cost.

Offline policy - Do not persist identifiable clinical data offline by default. Any future offline workflow requires explicit security, legal and clinical approval.

Specialized considerations

DICOM integration - Evaluate current OHIF and Cornerstone capabilities and licensing; integrate DICOMweb QIDO-RS/WADO-RS through authorized backend or PACS endpoints; never expose unrestricted PACS credentials to browsers.

Performance - Virtualize large worklists; lazy-load viewer; avoid fetching entire study into React state; memory and GPU limits tested on supported browsers; image rendering performance benchmarked with realistic de-identified studies.

Domain standards - FHIR ImagingStudy and DiagnosticReport mappings may be introduced through backend interoperability layer, not as assumptions about current APIs.